MASTERS COMPREHENSIVE EXAMINATION

Part I Theory (2 hours) 2 December 2017

Instructions: Attempt any three out of the following four questions on both pages. Tell us explicitly which three you would like to have graded. Open book and open notes.

1. Let X, Y be continuous random variables with joint pdf

$$f_{X,Y}(x,y) = \begin{cases} x+y & \text{if } 0<x<1,\ 0<y<1 \\ 0 & \text{otherwise.} \end{cases}$$

   a. Find the marginal density functions of X and Y.
   b. Find $Var(X)$.
   c. Find the $Cov(X,Y)$.
   d. Are X and Y independent? (Why?)
   e. Find the conditional density function of X given Y.
   f. Find $E(X|Y=1/2)$.

2. Let X_1, X_2, ... , X_n be i.i.d. continuous random variables each with density function

$$f(x|\theta) = \begin{cases} (\theta+1)x^{\theta} & \text{for } 0<x<1 \\ 0 & \text{otherwise.} \end{cases}$$

   a. Find the joint density function of X_1, X_2, ... , X_n.
   b. Find a sufficient statistic for θ.
   c. Find the maximum likelihood estimator of θ.

3. Suppose that the number X of side effects for a patient receiving a certain cancer treatment follows a Poison distribution with p.d.f. $p(x) = \lambda^x \exp(-\lambda)/x!$ for $x = 0, 1, 2, \ldots$, and $\lambda > 0$. Suppose that λ is a random variable with an exponential distribution with mean 1.
   a. Calculate the unconditional mean and variance of X.
   b. Calculate the unconditional discrete density of X.
   c. Repeat part b for the more general case when λ has an exponential distribution with mean q.

4. Suppose that X is an independent observation from a distribution with p.d.f. given by: $f(x) = 2\mu x - \mu + 1$, where $|\mu| \le 1$ and $0 < x < 1$.
   a. Construct the most powerful (MP) test of size α for testing the null hypothesis $H_0 : \mu = 0$ vs. the alternative $H_a : \mu = 1$.
   b. Construct the uniformly most powerful (UMP) test of size α for testing the null hypothesis $H_0 : \mu \le 0$ vs. the alternative $H_a : \mu > 0$.
   c. Suppose that the rejection region for testing $H_0 : \mu \le 0$ vs. the alternative $H_a : \mu > 0$ is $RR = \{X > 0.5\}$. What are the power and size of this test?